

Could FinTech Play a Role in Urban Planning?

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INTRODUCTION

Financial technologies (or FinTech) could change the way planners work. Distributed public ledgers and artificial intelligence (AI) used on big datasets represent disruptive technologies for public finance – if US financial regulators can keep up. FinTech could sweep aside existing local retail and commercial banks and their employees. They already have started. They will also sweep aside all kinds of existing public services. Private persons could invest in the tokens which might help fund generation alpha's electricity, gas, and even large-scale infrastructure like roads, dams and environmental clean-up services. Citizens acting in the decentralized autonomous organizations (DAOs) running payments could decide on local services, instead of in city council meetings. FinTech could do all this – and more.

Planners will need to adopt a flexible specialization mindset when planning for FinTech. Such a mindset accepts that service providers, not planners, will decide where local services sit. AI applications, using FinTech (rather than planning commissions and government economists), could help planners flexibly use licenses, permits and registrations to achieve voters' financial goals. Planners will thus need to understand FinTech in order to communicate its impacts to their constituents.

BACKGROUND

FinTech broadly applies to any online, digital technology which automates financial services. Types of technologies include:

- distributed public ledgers (DPLs) – these ledgers record the internet address and any related transactions like authorizing payments, payment amounts, or anything you can type into an Excel worksheet. They may sit on centralized servers in regulated banks or government agencies (centralized or on thousands/millions of users' computers (decentralized)).
- crypto-currencies (e.g. Bitcoin) – DPLs which generate a specific, encrypted address used online as currency, but which share the attributes of a security, commodity, insurance product, key, and more. Decentralized finance (or DeFi) refers to peer-to-peer (P2P) finance which skips banks. Central bank digital currencies (CBDCs) refer to regulated DPLs used as money.
- smart contracts – coded into cyptocurriences like Ethereum, these tokens self-execute if certain machine-verifiable conditions take place (like a resident making a tax payment). They will replace millions of lawyers, accountants, and other paper shifters in the future.
- tokens – a complex hyperlink to a DPL. These crypto-currency DPLs can have hard assets or debts behind them, making them securities. Many have emerged from protocols/platforms overseen by DAOs. Others like Bitcoin have emerged from large-scale collaborations.

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- artificial intelligence (like ChatGPT) – a type of machine learning involving millions of 1s or 0s which, when aggregated, act almost like the human brain. Machine learning allows platforms/apps to see complex patterns in large datasets.
- platforms/apps (e.g. Amazon or Venmo). They may offer services via webpages or via apps and will often have an ‘ecosystem’ of partners who provide related services. They may accept and use cryptocurrencies and tokens as payment or to interact with the platform.

FinTechs abroad have already offered tokens allowing citizens to pay for public toilets, electricity bills, and many other public services.

IMPACTS ON LOCAL FINANCIAL LANDSCAPES

Fintech will sweep away whole industries. FinTech already has created an employment boom and bust in San Francisco. BigTech firms struggling to use FinTech, like Apple, Google, and others have hired and fired thousands of IT engineers. In Miami, whole sections of the city have transformed thanks to crypto-currency traders. Few connect to FinTech the same changes happening in New York and other major US metros. Yet, the banking deserts, skid rows and big gentrification projects, which occur because of FinTech, have made national news. Atlanta (for AmEx), St. Louis (for Mastercard), and Charlotte (for Bank of America) represent examples of major metros currently undergoing radical change.

Preparing for this change, planners need a “flexible specialization” mindset when making plans. As such, planners should modularize – having banks, innovation centres and whole financial centres designed as removable, inter-changeable modules. They can also help explain these rapid changes to their constituencies (from tribal to national government). Planners can help them understand that large-scale changes occur because of changes in tastes and technologies and are not due to the political party in office or ‘capitalism.’ Planners can anticipate the large-scale privatization of public services and the revenue/expenditure impacts of such privatization and public-private partnerships in procurement.

BUDGETING AND PAYING FOR GOVERNMENT SERVICES

FinTech will actualize the e-government dream, if the SEC, Fed, FDIC, and other US regulators find a way to make **DeFi** a reality. FinTech will help governments (or their service providers) process decentralized payments, earn money on excess cash, borrow from their own populations (or abroad) and access tokenized online payment processing. DeFi could let planners’ constituents pay, lend, borrow and even mint payments for taxes and to pay for all government services (bypassing the Single Treasury Account) while still reporting to the OMB and Treasury. Inhabitants of smart cities can pay for services which come together funding and for them in flash mobs using their mobile phones. Planners thus need a flexible specialization mindset when planning for government offices which have remained in place for a century or longer.

Local **budgets** could reflect the same mindset. FinTech could tie property taxes to local needs more directly. Voters in DAOs could choose bands rather than a specific tax rate – with money leaving their digital wallets automatically. Residents could pay for business improvement district-related taxes, fees and fines – with the money going into a consolidation state-level or US-level digital wallet which reports to Treasury and the Fed. Online, dynamic financial statements could show real estate and property-related tax payments in real time. Financial technologies could tie such assessments to Zillow or other property valuation platforms

instantly. Disadvantaged groups could use them to fund housing for any one or more of their members, in effect mutualizing such demand – without the need for a potentially discriminatory banker getting involved.

DTL tech like **crypto-currencies** provides an unlimited number of actors with access to transactions as they occur. Local councils, departments of community affairs and other quasi-judicial planning-related groups can set up millions automatic conditions which, if met, will give permits and other authorizations to individuals and construction/design firms paying with tokens. Such smart contracts are baked into the currency which applicants may use. Social payments processors like Venmo, allow users to notify large groups when spending.

Such technologies will make public service demand more unstable. The Venmo equivalent of a ‘tweet’ can call up almost unlimited demand...or remove it. Governments may thus not need to rely on banks for their financial affairs. They could (in theory) even offer credit in buy-now pay-later schemes reminiscent of Affirm and UpStart.

FUNDING AND FINANCIAL REPORTING OF GOVERNMENT SERVICES

Tokens work like money, stocks, bonds derivatives and more. Thus, localities could bypass the large investment banks (like Goldman Sachs, Bancorp and others) when issuing general obligation and revenue bonds. Instead (if allowed), governments could plan on printing their own securitized tokens (which also serve as a currency) which give locals ownership stakes in local infrastructure and/or provides fixed income as desired. AI applications could keep the local asset register, participate in fixed-asset use, and automatically project the value of interest burdens and contingent liabilities.

FinTech would thus give planners a bigger role in **budgeting, receiving disbursements, and in financial reporting**. First, FinTech could make overruns obsolete. Second, planning bodies can tie current expenditure to revenue earned from fees and fines instantly. The token’s protocol could sweep any surpluses into interest-earning tokens, or tokens whose value they anticipate to appreciate. Imagine a fire station chief using a debit card whose account receives disbursements automatically when news of a fire appears online. Or who receives money when their DAO votes in these resources. Third, they can plan for a wide range of public services to move outside the public sector, thus freeing up land and staff costs. Fourth, FinTech can relieve much of the tedium experienced by accounting, legal and other remaining staff in local governments.

FLEXIBLE SPECIALIZATION IN CITY PLANS

Rapid social change has already challenged the social engineering theories of planning, from Ebenezer Howard to John Kasarda. FinTech-related technologies will only exacerbate these changes. Whole financial centers or cities – and their disaster mitigation activities -- could be planned and funded by private actors or AI.

Planning **deliberative bodies** (like planning councils) have a role to play when amending ordinances and other planning-related law. They can write FinTech right into ordinances, allowing them to self-execute when paid for. A modular, flexible specialization mindset could allow neighborhood groups to design and pay for their own plans. Smart contracts could accept such plans automatically if valid and paid for using tokens (as allowed by law). If not allowed, planners can lobby the US’s financial regulators to relax rules. Because the US falls behind places like the EU, the legality of such finance right now isn’t clear.

DAOs in particular could boost **democratic participation** in sub-national government. In such DAOs, local governments could poll citizens, allow them to set up their services, and even pay for them using the DAO's tokenized crypto-currency. They can also incorporate disaster-related planning and insurance into plans (described below). They can choose to use an existing pegged DAO (like MakerDAO) and its tokens (USDC), or make one of their own. Or they could wait for the Fed to issue a CBDC – thereby giving up the chance to use DeFi in the future. Local governments can pre-empt financial regulators now – by showing them the utility in issuing and using tokens to raise money, issuing tokenized debts, stuffing such tokens into automated derivatives-related DAOs, and arranging for mutual recognition of tokens issued in other jurisdictions.

They could tie spending to **equity** goals. Public school students' parents could use tokens to pay for the schools issuing their own tokens (and participate in their DAOs). They can tie local transfers to grades. Hospitals could similarly use tokens, instead of Wall Street, to raise money for current and capital spending. Local communities in DAOs could thus tie the subsidization of health care for blacks, Latinos and other disadvantaged groups, to pre-voted-on rules. Lenders could tie community development loans to a person's or company's zip code.

INCORPORATING DECENTRALIZED INVESTMENT AND INSURANCE IN PLANS

FinTech could change the way American sub-national governments work. They can completely privatize local services, infrastructure and contingent payments to token holders as well as issue debt. If allowed, such a privatization would move local capital investment budgets into private hands. They could tie debt with varying maturities to different tokens – allowing P2P lenders to governments to directly participate in the way these governments use their money. Such a move would reduce reliance on the type of collateralized debt obligations which caused the Global Financial Crisis. Right now, local governments mutualize these risks by making citizens pay, when bad luck strikes, in the form of higher local taxes.

DeFi-related FinTech has obvious impacts on the insurance which allows for disaster planning and **environmental protection**. Imagine a world where citizens bought pieces of earthquake-related risks. The token's protocol would debit their digital wallets when the quake occurred, and collect premia from them automatically if not. Citizens could decide, in the token's DAO how much to collect, what conditions merit a pay-out, and place restrictions on buying too many, correlated insurance-related tokens. Their crypto-currency's underlying protocol could also automatically harmonize EPA, state, county and city regulations and activity.

CONCLUSIONS

FinTech will likely replace many financial services-related jobs. FinTech will also affect many other jobs in government administration, hospital staffing, schools and even in fire stations. We have already seen these effects. Technology-related unemployment has produced many of the skid rows and urban no-go zones we see today. Crypto-currency related scams have pushed Congress to heavily regulate FinTech in the US. Such legislative clamp-downs are a mistake.

Because whole neighborhoods could appear or disappear overnight, planners need to adopt a flexible specialization mindset. Such a mindset allows them to slot in pre-designed modules of urban planning and associated budgets into their plans. AI programmes using automated FinTech applications could do some of this for them. Planners will play an important role in

explaining its benefits and drawbacks to their constituencies, using it when budgeting (and paying for plans), as well in moving the public-private boundary in service and infrastructure provision.

FinTech may offer planners new tools in fighting such unemployment, as well as in tackling environmental, equity and other issues. Tokens, and their related DAOs, could allow planners to act like bankers, increase democratic participation in government, and promote federal and sub-national coordination. AI could run whole services off FinTech-related protocols. Tokenized insurance could offer planners exciting new options in preventing, insuring against, and remedying future disasters. Planners will need to lobby national regulators to keep these activities legal.

This Note was prepared by Bryane Michael (AICP) with thanks from an unnamed colleague at the American Planning Association. A fuller version focusing on developing countries will be out soon.

FURTHER READING

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